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Omni-Scale Frequency Decomposition with Anatomy-Guided Edge Reconstruction for Camouflaged Human Detection in Defence Surveillance

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Abstract

Cross-border infiltration by personnel wearing ghillie suits and disruptive-pattern uniforms remains an unsolved surveillance problem along India's land borders: engineered concealment suppresses precisely the shape, edge and colour cues on which both human observers and conventional detectors depend. Research on camouflaged object detection has advanced rapidly, but its benchmarks are overwhelmingly animal-centric, and models trained on evolved biological camouflage transfer poorly to articulated, deliberately concealed humans. We address this gap on two fronts. First, we assemble a unified camouflaged-human benchmark of 4,170 images by harmonising four public corpora—ACD1K, CPD1K, MHCD2022, and a newly derived human-only subset of CAMO—under a common pixel-accurate mask, boundary, and pose-prior annotation scheme, with clip-aware splitting and cross-corpus perceptual-hash deduplication. Second, we propose OS-Res2Net-CHDNet, a 25.75 M-parameter network that couples a content-adaptive Feature Decomposition Module and a Spatial–Frequency Collaborative Attention block—which recover the residual high-frequency texture discontinuity that camouflage cannot fully erase—with an OSNet-style omni-scale neck, an Anatomy-guided Edge Reconstruction module that injects a 17-keypoint pose prior additively, and an image-level presence gate that suppresses false alarms on empty frames. On the combined test split the model attains $S_\alpha = 0.XXX$, MAE $= 0.0XX$, and $F^{\text{bd}} = 0.XXX$. The resulting system is intended to strengthen the automated surveillance capability of the Indian armed forces.

Keywords: Camouflaged human detection; Camouflaged object detection; Defence surveillance; Frequency-domain feature decomposition; Omni-scale feature learning; Pose-guided attention; Border infiltration

1. Introduction

Infiltration across India's northern and western land frontiers is a persistent and quantitatively documented security burden. Official data placed before Parliament by the Ministry of Home Affairs record net infiltration into Jammu

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and Kashmir falling from 141 in 2019 to 51 in 2020, 34 in 2021, and 14 in 2022 [1]; independent longitudinal tracking by the South Asia Terrorism Portal reports the same downward trajectory, together with the launch-pad and training-camp infrastructure that sustains it [2]. The Ministry of Defence Year-End Review for 2022 records twelve infiltration attempts neutralised along the Line of Control (LoC), in which eighteen foreign terrorists were eliminated and substantial quantities of arms and ammunition recovered [3]; press reporting for 2023 indicates a partial resurgence, with roughly forty infiltrators killed while attempting to cross [4]. The operationally decisive quantity, however, is not the declining trend but the residual non-zero count: the LoC extends for approximately 740 km across densely forested, boulder-strewn, and seasonally snowbound terrain, and every crossing that succeeds is, by definition, a failure of the surveillance grid rather than of the interception force. Because interdiction is contingent on detection, the marginal return on improving automated detection at the sensor is substantially higher than on adding further downstream response capacity.

That detection problem is hard for reasons that are physical rather than procedural. Ghillie suits, disruptive-pattern uniforms, and locally improvised vegetation garnish are engineered specifically to defeat the visual cues that segmentation depends on: they fracture the body outline, randomise the internal texture statistics, and drive the foreground colour distribution towards that of the background, so the figure–ground boundary on which both an observer and a detector rely is deliberately erased. Comparative work in camouflage science establishes that disruptive coloration is effective precisely because it interferes with edge assignment and with the observer's acquired search image, and that this interference persists even when the target is fixated [5, 6]. The failure is not confined to the visible band: security agencies along the LoC have reported the use of thermal-camouflage suits that insulate the body's heat signature sufficiently that hand-held thermal imagers could not resolve the silhouette they normally rely on [7], which removes the assumption that an infrared channel provides a camouflage-invariant fallback. Compounding this, sustained monitoring of low-event-rate video imposes a well-documented vigilance decrement on human operators, whose detection sensitivity degrades measurably within the first half hour of watch [8]. Our own measurements over the corpora assembled in this work quantify how extreme the resulting regime is: mean foreground occupancy is 0.0199 on CPD1K and 0.0171 on MHCD, meaning the target typically occupies under two percent of the frame, while the mean local camouflage contrast—the intensity separation between the target and an annulus of its immediate surround—falls as low as 0.0675 on CAMO-Human. At that separation the figure–ground signal approaches the sensor noise floor, and detection cannot be recovered by contrast enhancement alone.

Camouflaged object detection (COD) has matured rapidly as a research area, but almost entirely on animal subjects. The field's standard benchmarks—CAMO [9], CHAMELEON [10], COD10K [11], and NC4K [12]—are dominated by biologically camouflaged fauna, and the methods trained on them inherit that bias, from the search-and-identification paradigm of SINet and SINet-v2 [11, 13] through distraction mining [14], mutual graph learning [15], explicit boundary guidance [16], frequency-domain analysis [17], feature decomposition with edge reconstruction [18], texture-aware refinement [19], high-resolution iterative feedback [20], masked separable attention [21], mixed-scale pyramids [22], gradient supervision [23], and edge–semantic collaboration [24]. The transfer to camouflaged humans is weaker than the surface similarity of the tasks suggests. Biological camouflage is evolved, statistically stationary within a species, and worn on a body plan whose articulation is limited; military concealment is deliberately engineered, changes with issue and theatre, and covers a highly articulated skeleton that adopts prone, crouched, and partially occluded postures specifically to break the silhouette a detector would key on. The human-specific literature that does exist remains small and mutually isolated: CPD1K established the first camouflaged-people corpus and a dense deconvolution baseline over woodland and snowfield scenes [25], MHCD2022 introduced a military high-level camouflage benchmark with an accompanying detection network [26], and ACD1K contributed high-resolution imagery of ghillie-suited and tactically equipped individuals across desert, forest, and snow [27]. Each is small, each uses its own annotation convention and split protocol, and no unified, deduplicated, human-only benchmark spans them.

Three technical gaps follow directly from this characterisation, and they organise our design. First, camouflage is engineered to destroy *spatial* contrast, but the fabric-to-vegetation transition still leaves a residual discontinuity in high-frequency texture statistics that no dyeing process fully removes; a detector that operates only on the spatial-domain feature map discards this evidence. We therefore decompose every backbone feature into an adaptively estimated low-frequency component and its high-frequency residual, and recombine them under learned per-location gates, so the network can weight the surviving texture cue exactly where the spatial cue has collapsed. Second, camouflaged human targets span an extreme scale range—from a distant prone figure a few dozen pixels across to a partially occluded torso filling the frame—which a fixed receptive field cannot serve; we address this with an omni-scale neck of parallel multi-depth streams fused by a shared aggregation gate, following the omni-scale design principle established for person re-identification [29], over a Res2Net backbone that already carries granular multi-scale structure within each residual block [28]. Third, humans possess a skeletal prior that no animal-centric model has reason to exploit, and a surveillance deployment must additionally be able to answer "nothing here" rather than hallucinating a target on every empty frame. We inject a precomputed 17-keypoint pose prior [34] through an anatomy-guided attention module whose gating is strictly additive, so a missed keypoint can never suppress a genuine detection, and we gate the final mask by an image-level presence classifier in the manner of the anabranch design [9].

The contributions of this work are as follows.

- A unified camouflaged-human benchmark of 4,170 images harmonising ACD1K, CPD1K, MHCD2022, and a newly derived human-only subset of CAMO under a single annotation scheme comprising binary masks, 4-pixel boundary maps, pose priors, and image-level presence labels, with clip-aware splitting and cross-corpus perceptual-hash deduplication.
- A content-adaptive Feature Decomposition Module and a Spatial–Frequency Collaborative Attention block that jointly recover the high-frequency texture evidence camouflage leaves behind, without requiring an explicit wavelet or Fourier transform.
- An Anatomy-guided Edge Reconstruction module that conditions segmentation on a 17-keypoint pose prior through a purely additive gate, together with a presence-gated output that suppresses false alarms on target-free frames.
- OS-Res2Net-CHDNet, a 25.75 M-parameter architecture integrating the above, evaluated against ten representative COD baselines across four camouflaged-human datasets under five standard metrics.

2. Related Work

3. Preliminaries and Research Methodology

3.1. Overview of the research pipeline

The study proceeds in five stages. *(i) Corpus harmonisation*: four heterogeneous public sources, annotated variously with pixel masks, JPEG-compressed grey-scale ground truth, and bounding boxes only, are converted to a single on-disk representation of RGB image, strictly binary mask, binary boundary map, pose prior, and image-level presence flag. *(ii) Prior precomputation*: a pose estimator is run once over the harmonised corpus and its keypoint heat maps are cached, so that the anatomical prior costs nothing at training time and the segmentation network never backpropagates into the pose model. *(iii) Split construction*: clip-aware partitioning prevents temporally adjacent frames of the same subject from straddling a split, and a perceptual-hash pass removes cross-corpus near duplicates that would otherwise leak test content into training. *(iv) Training*: OS-Res2Net-CHDNet is optimised under a four-term objective combining a boundary-weighted structure loss, deep supervision, edge supervision, and presence classification, with multi-scale sampling. *(v) Presence-gated inference and evaluation*: a single forward pass yields a continuous saliency map gated by the predicted presence probability, scored under five standard COD metrics. Figure 1 summarises the network stage of this pipeline.

3.2. Problem formulation

Let $I \in \mathbb{R}^{3\times H\times W}$ be an input frame and $P \in \mathbb{R}^{17\times H_p\times W_p}$ its cached pose prior. The task is to predict a per-pixel probability map $\hat{Y} \in [0,1]^{1\times H\times W}$ indicating camouflaged human presence, supervised by a binary mask $M \in \{0,1\}^{1\times H\times W}$ and a boundary map $B \in \{0,1\}^{1\times H\times W}$. Because the deployment must tolerate target-free frames, each sample additionally carries a scalar presence label $y \in \{0,1\}$, with $y = 0$ for negatives and $M \equiv 0$ in that case. Throughout, $\sigma(\cdot)$ denotes the sigmoid, $\delta(\cdot)$ the ReLU, $\odot$ the Hadamard product, $[\cdot\,;\cdot]$ channel concatenation, and $\mathrm{GAP}(\cdot)$ global average pooling.

[Figure placeholder — end-to-end architecture of OS-Res2Net-CHDNet]

Input → Res2Net-50 encoder ($F_1 \ldots F_4$) → 1×1 channel reduction → per-level FDM (low/high-frequency split) → SFA (gated recombination) → omni-scale neck → AER (pose-gated) → partial decoder → {side masks, edge map, presence gate} → presence-gated output mask.

Fig. 1. End-to-end architecture of OS-Res2Net-CHDNet. The four pyramid levels are processed by independent instances of the FDM, SFA, omni-scale, and AER modules before top-down multiplicative decoding.

3.3. Multi-scale encoder

The encoder is Res2Net-50 (`res2net50_26w_4s`) [28], pre-trained on ImageNet [32]. It is chosen over a plain residual network because its hierarchical residual-like connections express multiple receptive fields *within* a single block, which matters when the discriminative evidence is a texture-statistic mismatch at an unknown spatial frequency. Four pyramid levels are taken at strides $4, 8, 16, 32$ with 256, 512, 1024, and 2048 channels, and each is projected to a common width $C = 64$:

$$F_i = \delta\Big(\mathrm{BN}\Big(\mathrm{Conv}_{1\times 1}(F_i^{\mathrm{bb}})\Big)\Big), \qquad i = 1, \ldots, 4. \tag{1}$$

Every subsequent module is applied independently at all four levels and is fully resolution-agnostic, so the input size is a data-loading choice rather than an architectural constant.

3.4. Feature Decomposition Module

The FDM separates each level into a smooth regional component and the high-frequency residual that carries the surviving texture evidence. Rather than fixing a cut-off, the module estimates it per image from global context. Three average-pooled estimates at kernel sizes $k \in \{3, 5, 7\}$ are computed with unit stride,

$$S_k = \mathrm{AvgPool}_k(F), \qquad k \in \{3, 5, 7\}, \tag{2}$$

and combined under weights predicted from the channel descriptor of F:

$$\boldsymbol{\alpha} = \mathrm{softmax}(W_2\, \delta(W_1\, \mathrm{GAP}(F))) \in \mathbb{R}^3, \qquad W_1 \in \mathbb{R}^{16\times C},\ W_2 \in \mathbb{R}^{3\times 16}. \tag{3}$$

The low-frequency estimate and its residual are then

$$F_{\mathrm{LF}} = \alpha_3 S_3 + \alpha_5 S_5 + \alpha_7 S_7, \qquad F_{\mathrm{HF}} = F - F_{\mathrm{LF}}. \tag{4}$$

The residual is sparse and high-variance, so it is stabilised by group normalisation before a depthwise–pointwise refinement:

$$\hat{F}_{\mathrm{HF}} = \delta(\mathrm{BN}(\mathrm{Conv}_{1\times1}(\delta(\mathrm{BN}(\mathrm{DWConv}_{3\times3}(\mathrm{GN}_8(F_{\mathrm{HF}}))))))) . \tag{5}$$

Making α input-dependent is what distinguishes this from a fixed band-pass filter: a distant prone target in dense woodland and a near-field figure against snow place their discriminative energy at different scales, and a single learned cut-off cannot serve both.

3.5. Spatial–Frequency Collaborative Attention

The two bands are not uniformly informative across an image: in a well-blended region the high-frequency residual dominates, whereas at an occlusion boundary the low-frequency component is more reliable. The SFA block therefore learns a spatially varying recombination. Writing $\Phi = [F_{\mathrm{LF}}; \hat{F}_{\mathrm{HF}}] \in \mathbb{R}^{2C\times H\times W}$, two independent gates are produced,

$$W_{\mathrm{LF}} = \sigma\Big(\mathrm{Conv}^{(a)}_{3\times3}(\Phi)\Big), \qquad W_{\mathrm{HF}} = \sigma\Big(\mathrm{Conv}^{(b)}_{3\times3}(\Phi)\Big), \tag{6}$$

and applied to yield the fused response and a residual base,

$$F_{\mathrm{fuse}} = W_{\mathrm{LF}} \odot F_{\mathrm{LF}} + W_{\mathrm{HF}} \odot \hat{F}_{\mathrm{HF}}, \qquad F_{\mathrm{base}} = F + F_{\mathrm{fuse}}. \tag{7}$$

A self-refinement gate then sharpens the response:

$$F_{\mathrm{ref}} = \sigma(\mathrm{Conv}_{3\times3}(\delta(\mathrm{BN}(\mathrm{Conv}_{3\times3}(F_{\mathrm{base}}))))), \qquad F_{\mathrm{SFA}} = F_{\mathrm{base}} + F_{\mathrm{base}} \odot F_{\mathrm{ref}}. \tag{8}$$

Note that the gates are independent rather than a softmax pair, so the block may attenuate or amplify both bands at a location instead of being forced to trade one against the other.

3.6. Omni-scale neck

Each level is then passed through an omni-scale block adapted from OSNet [29]. The input is reduced to a bottle-neck width C/r with $r = 4$ and processed by $T = 4$ parallel streams built from stacked lightweight units,

$$\mathrm{Lite}_{3\times3}(u) = \delta(\mathrm{BN}(\mathrm{DWConv}_{3\times3}(\mathrm{Conv}_{1\times1}(u)))), \qquad T_t = \underbrace{\mathrm{Lite}_{3\times3} \circ \cdots \circ \mathrm{Lite}_{3\times3}}_{t \text{ times}}(x'), \tag{9}$$

so that stream t has an effective receptive field of $2t + 1$, giving the set $\{3, 5, 7, 9\}$. A single aggregation gate, shared across streams so that their weights remain mutually calibrated, produces input-dependent channel weights,

$$g_t = \sigma(W_4\, \delta(W_3\, \mathrm{GAP}(T_t))), \qquad y = \sum_{t=1}^{T} g_t \odot T_t, \tag{10}$$

and the block closes with expansion and a residual connection:

$$F_{\mathrm{OS}} = \delta(\mathrm{BN}(\mathrm{Conv}_{1\times1}(y)) + x)\,. \tag{11}$$

Sharing the gate is deliberate: with per-stream gates the block degenerates towards independent scale-specific branches, whereas a shared gate forces a single input-conditioned decision about which scale carries the evidence in this image.

3.7. Anatomy-guided Edge Reconstruction

The AER module supplies the human-specific prior. A cached pose tensor of 17 COCO keypoint heat maps [34] is resampled to the level's resolution, concatenated with the feature, and mapped to a single-channel spatial attention map:

$$P' = \mathrm{Resize}(P, H \times W), \qquad A = \sigma(\mathrm{Conv}_{3\times3}(\delta(\mathrm{BN}\,(\mathrm{Conv}_{3\times3}([P'; F_{\mathrm{OS}}])))))\,. \tag{12}$$

The gate is applied additively rather than multiplicatively:

$$\tilde{F} = F_{\mathrm{OS}} \odot (A + 1). \tag{13}$$

The unit offset in Eq. (13) is the critical design choice. A conventional multiplicative gate $F \odot A$ would allow the pose branch to *veto* a location, which is exactly the wrong failure mode here: the pose estimator is itself operating on camouflaged input and will frequently return no keypoints for precisely the hardest targets. With the offset, a location carrying zero anatomical evidence passes through unchanged, so the prior can only add emphasis and never censor a detection the segmentation head believes in. The module is thus safe to apply even when the pose channel is entirely uninformative.

3.8. Partial decoder and multi-task heads

Decoding follows a top-down multiplicative cascade in the spirit of the partial decoder [30], in which coarse semantic levels gate finer ones rather than being summed with them:

$$x_4 = \tilde{F}_4, \qquad x_i = \tilde{F}_i \odot \mathrm{Up}(\mathrm{Conv}_{1\times1}(x_{i+1}))\,, \quad i = 3, 2, 1. \tag{14}$$

All four levels are resampled to the resolution of x_1, concatenated, and refined to the main logit:

$$Z = \mathrm{Conv}_{1\times1}\Big(\varphi(\varphi([\,\mathrm{Up}(x_4); \mathrm{Up}(x_3); \mathrm{Up}(x_2); x_1\,]))\Big), \qquad \varphi(\cdot) = \delta(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\cdot))). \tag{15}$$

Three head families are attached. Deep supervision applies a 1×1 logit to each decoder level, $S_i = \mathrm{Up}(\mathrm{Conv}_{1\times1}(x_i))$; an edge head predicts the boundary map from the finest level, $E = \mathrm{Up}(\mathrm{Conv}_{1\times1}(x_1))$; and an anabranch-style presence gate [9] classifies the frame from the deepest raw encoder feature,

$$p = W_6\,\mathrm{Drop}_{0.3}\Big(\delta\Big(W_5\,\mathrm{GAP}(F_4^{\mathrm{bb}})\Big)\Big)\,, \qquad W_5 \in \mathbb{R}^{256\times2048},\ W_6 \in \mathbb{R}^{1\times256}. \tag{16}$$

The prediction is the presence-gated mask,

$$\hat{Y} = \sigma(Z) \cdot \sigma(p), \tag{17}$$

so a confident negative multiplicatively suppresses the entire map. This matters operationally far more than it does on a standard COD benchmark, where every test image contains a target by construction; a border surveillance feed is overwhelmingly target-free, and an ungated model accumulates false alarms at the frame rate of the sensor.

3.9. Training objective

Boundaries dominate the difficulty of this task, so pixels are weighted by their local boundary proximity,

$$\omega = 1 + 5 \left| \mathrm{AvgPool}_{31}(M) - M \right|, \tag{18}$$

which is near unity in flat interior and exterior regions and rises sharply in the transition band. The structure loss combines a weighted cross-entropy with a weighted soft IoU. With $\pi = \sigma(\cdot)$ the predicted probability,

$$\mathcal{L}_{\text{wbce}} = \frac{\sum \omega \cdot \mathrm{BCE}(\pi, M)}{\sum \omega}, \qquad \mathcal{L}_{\text{wiou}} = 1 - \frac{\sum \omega \pi M + 1}{\sum \omega(\pi + M) - \sum \omega \pi M + 1}, \tag{19}$$

$$\mathcal{L}_{\text{str}}(\cdot) = \mathcal{L}_{\text{wbce}} + \mathcal{L}_{\text{wiou}}, \tag{20}$$

following the structure-loss formulation established for saliency and adopted throughout COD [31]. Boundary pixels are of the order of 0.1–1% of the frame, so the edge term pairs cross-entropy with a Dice term,

$$\mathcal{L}_{\text{dice}}(\pi, B) = 1 - \frac{2 \sum \pi B + 1}{\sum \pi + \sum B + 1}, \qquad \mathcal{L}_{\text{edge}} = \frac{\sum_n y_n \left[\mathrm{BCE}(\pi_n, B_n) + \mathcal{L}_{\text{dice}}(\pi_n, B_n)\right]}{\sum_n y_n}, \tag{21}$$

where the masking by y_n excludes negatives, which have no boundary to supervise, while preserving a static batch shape. The presence term is a plain cross-entropy, $\mathcal{L}_{\text{pres}} = \mathrm{BCE}(\sigma(p), y)$. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{str}}(Z) + \sum_{i=1}^{4} \lambda_i \, \mathcal{L}_{\text{str}}(S_i) + \lambda_e \mathcal{L}_{\text{edge}} + \lambda_p \mathcal{L}_{\text{pres}}, \tag{22}$$

with $\boldsymbol{\lambda} = (0.4, 0.6, 0.8, 1.0)$ ordered coarse to fine, $\lambda_e = 1.0$, and $\lambda_p = 0.5$. The ascending side weights reflect that the finest decoder level is closest to the supervised output resolution, while the coarse levels act mainly as a localisation regulariser.

3.10. Implementation details

Table 1 lists the full configuration. Optimisation uses AdamW [33] with a discriminative learning rate: the ImageNet-initialised encoder is trained at one tenth the rate of the randomly initialised modules, and its batch-normalisation statistics are frozen for the first five epochs so that the untrained heads cannot corrupt the pre-trained representation while their gradients are large. Geometric augmentation is applied by a single shared affine warp so that image, mask, boundary map, and all seventeen pose channels remain in exact correspondence; photometric jitter is applied to the image only.

Table 1. Training configuration for OS-Res2Net-CHDNet.

Group	**Setting**	**Value**
Architecture	Encoder	Res2Net-50 (`res2net50_26w_4s`), ImageNet pre-trained
	Common feature width C	64
	Omni-scale streams T	4 (receptive fields 3, 5, 7, 9)
	Pose prior	17 COCO keypoints, cached at $H/4 \times W/4$
	Total parameters	25.75 M
Optimisation	Optimiser	AdamW, weight decay 1×10^{-4}
	Base learning rate	1×10^{-4}
	Encoder learning-rate multiplier	0.1
	Schedule	cosine annealing, 5-epoch linear warm-up
	Epochs / batch size	100 / 16
	Gradient clipping (global norm)	0.5
	Encoder BatchNorm frozen	first 5 epochs
	Mixed precision / weight EMA	enabled / decay 0.999
Data	Input resolution	352×352
	Multi-scale sampling	{0.75, 1.00, 1.25}, one scale per step
	Horizontal flip	$p = 0.5$
	Rotation / scale jitter	±15° / [0.75, 1.25]
	Colour jitter (image only)	0.2 brightness, contrast, saturation
Protocol	Random seed	42
	Model selection	best validation S_α
	Inference	single forward pass, no test-time augmentation

3.11. Novelty and expected impact

Three properties distinguish this design from the COD literature it builds on. First, frequency-domain reasoning in COD has previously relied on explicit transforms—discrete cosine analysis [17] or learnable wavelets [18]—whereas the FDM obtains an equivalent band separation from adaptively weighted pooling, at negligible cost and with a cut-off that is conditioned on image content rather than fixed at design time. Second, prior work injects boundary or gradient priors that are derived from the mask itself [16, 23, 18]; the AER instead injects an *external, semantically typed* prior specific to the human body plan, and does so through a strictly additive gate that cannot degrade performance when the prior is absent—a property that matters precisely because pose estimation fails hardest on the same inputs segmentation finds hardest. Third, the presence gate targets an operating point the standard COD protocol never measures. Benchmarks guarantee a target in every image, so a model that never predicts emptiness is never penalised; a border surveillance feed inverts that ratio, and the false-alarm rate on target-free frames determines whether an operator trusts the system at all. Retaining 1,024 animal images from CAMO as explicit negatives, rather than discarding them once the human subset was extracted, supplies the supervision this gate requires.

4. Dataset

4.1. Constituent corpora

Four public sources are combined. **ACD1K** [27] contributes 1,077 high-resolution images of individuals in ghillie suits and modern tactical outdoor gear across desert, rocky, forest, and snow terrain; its ground truth is distributed as JPEG-compressed grey-scale, which introduces a small band of ambiguous intensities that we binarise at 127 and measure to affect 0.11% of pixels on average. **CPD1K** [25] contributes 1,000 frames drawn from 260 video clips in woodland and snowfield scenes, carrying pattern-family labels: woodland (499), desert (150), digital (150), none (101), snow (50), and urban (50). **MHCD2022** [26] contributes military camouflage imagery annotated only with bounding boxes over five object categories, of which we retain the person class. **CAMO-Human** is our own derivation: CAMO [9] contains both animal and human camouflage without a class label separating them, so we extract the human subset automatically and retain the remainder as negatives.

4.2. Annotation harmonisation

The four sources arrive in incompatible annotation formats, and reconciling them is a substantive part of this contribution. For MHCD, which ships boxes only, we generate pixel masks by prompting the Segment Anything family of models [35, 36, 37] with each person box, followed by automated per-box quality control that rejects masks leaking outside the prompt box, occupying an implausible fraction of it, or fragmenting into disconnected components; of 3,364 prompted boxes, 2,757 were accepted, 544 rejected for leakage, 59 for excessive area, and 4 for fragmentation or an undersized box. For CAMO, the human subset is identified by scoring a CLIP [38] embedding of the ground-truth object crop against human and animal text prompts, cross-checked with a person detector, and confirmed by manual review of the contact sheet; this yields 226 human images and retains 1,024 animal images as presence-gate negatives. Across all corpora, boundary ground truth is derived from each mask by morphological erosion and dilation to a 4-pixel band, and pose priors are precomputed once with a keypoint estimator [39], stored as 17-channel half-precision heat maps at quarter resolution with per-keypoint confidence weighting; frames in which no person is detected receive an all-zero prior, which Eq. (13) handles without special casing.

4.3. Splits, deduplication, and leak repair

Splits are constructed to eliminate three distinct leakage channels. Temporal leakage is addressed for CPD1K by partitioning at the level of its 260 source clips rather than individual frames, so that no clip straddles a split boundary. Cross-corpus leakage is addressed by a perceptual difference-hash pass over the union, which identified 1,283 near-duplicate pairs and removed 611 images, predominantly between MHCD and CPD1K. Finally, the MHCD distribution as shipped defines a validation set that is a subset of its training set; we detected and repaired this during preparation, since leaving it would have made validation-based model selection meaningless. Table 2 reports the resulting composition.

Table 2. Composition of the unified camouflaged-human benchmark after quality control and deduplication.

Corpus	Retained	Negatives	Train	Val	Test	Original annotation
ACD1K [27]	1,077	0	673	75	329	grey-scale mask (JPEG)
CPD1K [25]	1,000	0	657	128	215	binary mask, 260 clips
CAMO-Human [9]	226	1,024	100	14	112	mask, class not labelled
MHCD2022 [26]	2,478	376	1,592	392	494	bounding box only
Combined	**4,170**	**1,400**	**2,523**	**497**	**1,150**	harmonised

[Figure placeholder — benchmark composition and difficulty]

(a) retained versus excluded images per corpus; (b) train/validation/test split sizes; (c) foreground-area distribution per corpus.

Fig. 2. Composition and difficulty profile of the unified camouflaged-human benchmark.

4.4. *Difficulty characterisation*

Table 3 quantifies why this benchmark is hard, and shows that its four constituents are hard in different ways. CPD1K and MHCD are dominated by small, distant targets, with mean foreground occupancy below 2% of the frame; ACD1K and CAMO-Human contain far larger targets but blend them far more effectively, CAMO-Human recording the lowest local camouflage contrast in the collection at 0.0675. A model tuned only for small-object recall would therefore succeed on two corpora and fail on the other two, which is the principal argument for evaluating on the combination rather than on any single source.

Table 3. Difficulty statistics per corpus. Foreground area is the fraction of pixels labelled target; local camouflage contrast measures intensity separation between the target and its immediate surround, with lower values indicating better blending.

Corpus	**Foreground area**				**Camouflage contrast**
	mean	median	p_{10}	p_{90}	
ACD1K	0.1850	0.1619	0.0321	0.3902	0.0954
CPD1K	0.0199	0.0100	0.0033	0.0479	0.1172
CAMO-Human	0.1962	0.1571	0.0438	0.4086	0.0675
MHCD2022	0.0171	0.0055	0.0018	0.0352	0.1035

5. Experimental Study

5.1. *Evaluation metrics*

Five metrics are reported, following standard COD practice. Mean absolute error measures the pixel-wise deviation of the continuous prediction $\hat{Y}$ from the ground truth M,

$$\mathrm{MAE} = \frac{1}{HW} \sum_{u=1}^{H} \sum_{v=1}^{W} \left| \hat{Y}(u, v) - M(u, v) \right| . \tag{23}$$

The F-measure balances precision and recall with a recall-discounting factor $\beta^2 = 0.3$ [42],

$$F_\beta = \frac{(1 + \beta^2)\,\mathrm{Precision} \cdot \mathrm{Recall}}{\beta^2\,\mathrm{Precision} + \mathrm{Recall}}, \tag{24}$$

evaluated over 255 thresholds and reported at the adaptive threshold $\min(1, 2\bar{\hat{Y}})$. The structure measure [40] combines object-aware and region-aware structural similarity with $\alpha = 0.5$,

$$S_\alpha = \alpha\, S_o + (1 - \alpha)\, S_r, \tag{25}$$

and the enhanced-alignment measure [41] captures joint pixel-level and image-level agreement through the enhanced alignment matrix ϕ,

$$E_\phi = \frac{1}{HW} \sum_{u=1}^{H} \sum_{v=1}^{W} \phi(u, v). \tag{26}$$

Because boundary fidelity is the property the architecture is explicitly optimised for, we additionally report a boundary F-measure F^{bd} computed over mask contours with a matching tolerance of 0.75% of the image diagonal.

5.2. Experimental setup

Each model is trained independently on each corpus and on the combination, under the configuration of Table 1, and evaluated on the corresponding held-out test split. Baselines are re-trained on our splits rather than quoted from their original papers, since the corpora, the deduplication, and the leak repair all differ from the published protocols; quoting original numbers against our splits would not be a valid comparison. All methods use their published backbone and default hyperparameters.

5.3. Comparison with state-of-the-art methods

Tables 4 and 5 report the comparison across the four corpora, arranged as two vertical blocks of two datasets each.

Table 4. Quantitative comparison on ACD1K and CAMO-Human. Best results in bold. All values are placeholders pending training.

Method	**ACD1K**					**CAMO-Human**				
	MAE↓	F_β ↑	S_α ↑	E_ϕ ↑	F^{bd} ↑	MAE↓	F_β ↑	S_α ↑	E_ϕ ↑	F^{bd} ↑
SINet-v2 [13]	–	–	–	–	–	–	–	–	–	–
DDCN [25]	–	–	–	–	–	–	–	–	–	–
TANet [19]	–	–	–	–	–	–	–	–	–	–
FEDER [18]	–	–	–	–	–	–	–	–	–	–
HitNet [20]	–	–	–	–	–	–	–	–	–	–
CamoFormer [21]	–	–	–	–	–	–	–	–	–	–
ZoomNeXt [22]	–	–	–	–	–	–	–	–	–	–
ESCNet [24]	–	–	–	–	–	–	–	–	–	–
DGNet [23]	–	–	–	–	–	–	–	–	–	–
Ours	**0.0XX**	**0.XXX**	**0.XXX**	**0.XXX**	**0.XXX**	**0.0XX**	**0.XXX**	**0.XXX**	**0.XXX**	**0.XXX**

5.4. Ablation study

Table 6 isolates the contribution of each module by cumulative addition to a Res2Net-50 encoder with the partial decoder alone. The presence-gate row is evaluated on the combined split including its 1,400 negatives, since a gate cannot be assessed on a benchmark without target-free frames.

Table 5. Quantitative comparison on CPD1K and MHCD2022. Best results in bold. All values are placeholders pending training.

Method	**CPD1K**					**MHCD2022**				
	MAE↓	F_β ↑	S_α ↑	E_ϕ ↑	F^{bd} ↑	MAE↓	F_β ↑	S_α ↑	E_ϕ ↑	F^{bd} ↑
SINet-v2 [13]	–	–	–	–	–	–	–	–	–	–
DDCN [25]	–	–	–	–	–	–	–	–	–	–
TANet [19]	–	–	–	–	–	–	–	–	–	–
FEDER [18]	–	–	–	–	–	–	–	–	–	–
HitNet [20]	–	–	–	–	–	–	–	–	–	–
CamoFormer [21]	–	–	–	–	–	–	–	–	–	–
ZoomNeXt [22]	–	–	–	–	–	–	–	–	–	–
ESCNet [24]	–	–	–	–	–	–	–	–	–	–
DGNet [23]	–	–	–	–	–	–	–	–	–	–
Ours	**0.0XX**	**0.XXX**	**0.XXX**	**0.XXX**	**0.XXX**	**0.0XX**	**0.XXX**	**0.XXX**	**0.XXX**	**0.XXX**

Table 6. Cumulative ablation on the combined test split. All values are placeholders pending training.

#	**Configuration**	**Params (M)**	MAE↓	F_β ↑	S_α ↑	F^{bd} ↑
1	Res2Net-50 + partial decoder (baseline)	–	–	–	–	–
2	+ FDM	–	–	–	–	–
3	+ SFA	–	–	–	–	–
4	+ omni-scale neck	–	–	–	–	–
5	+ AER (pose prior)	–	–	–	–	–
6	+ presence gate (full model)	25.75	0.0XX	–	0.XXX	0.XXX

5.5. Qualitative results

[Figure placeholder — qualitative comparison]

Columns: input, ground truth, SINet-v2, FEDER, ZoomNeXt, Ours.
Rows: prone woodland target, snowfield target, desert ghillie suit, and a target-free negative frame.

Fig. 3. Qualitative comparison on representative hard cases from each corpus, including a target-free frame illustrating the effect of the presence gate.

6. Conclusion

This work treats camouflaged human detection as a problem distinct from camouflaged object detection rather than a special case of it, and addresses it on both the data and the model side. We assembled a unified benchmark of 4,170 images from four previously isolated corpora under a single annotation scheme, with clip-aware splitting, cross-corpus perceptual-hash deduplication, and repair of a train/validation leak present in one of the source distributions. On that benchmark we introduced OS-Res2Net-CHDNet, which recovers the high-frequency texture evidence that engineered camouflage leaves behind through content-adaptive frequency decomposition and gated spatial–frequency recombination, aggregates it across an omni-scale receptive-field bank, conditions it on a 17-keypoint anatomical prior through a strictly additive gate, and suppresses false alarms on target-free frames through an image-level presence gate.

Several limitations bound these results. The MHCD masks are SAM-derived pseudo-labels rather than human annotations, and although automated quality control rejected roughly eighteen percent of prompted boxes, residual boundary noise remains and will slightly depress the boundary F-measure on that corpus. The CAMO-Human subset comprises only 226 images, which limits what can be concluded from it in isolation. Most importantly, the system operates on visible-band imagery alone, whereas the reported use of thermal-camouflage garments along the LoC [7] indicates that a deployed capability will eventually require multi-spectral fusion. Future work will therefore extend the benchmark to paired visible and thermal imagery, replace the SAM pseudo-labels on MHCD with verified annotations, and evaluate the presence gate against the sustained false-alarm rate on continuous target-free footage, which is the metric that ultimately determines operator trust in a deployed surveillance system.

Acknowledgements

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